LedgerLens

Root-cause analysis as a set intersection over a ledger of business changes, verified by negative controls.

Dashboards tell you *what* moved. Finding out *why* takes an analyst days across Slack, Jira and Zendesk — and the expensive failure mode is acting on the wrong why (cutting marketing spend when checkout was broken). LedgerLens takes an anomalous metric, narrows it to the cohort that actually moved, intersects that cohort with every recorded change in the business, and then tries to *kill* each candidate with automatically generated negative controls. What survives is ranked, and every number on screen links to the SQL that produced it.

Built for the Accenture Innovation Challenge 2026, problem statement 3.

> **New here?** `docs/how_it_works.md` explains the whole system from zero -- no analytics background assumed. Everything below assumes the vocabulary it teaches.
>
> **Want the full ordered path through the repo?** `study_guide.md` sequences every document and source file end to end, with checkpoints.

Read this first: what this does and does not claim

It does not prove causation, and it does not try to. With a single incident there is no population to estimate an effect over, so any tool claiming "causal inference" here is overselling. What LedgerLens claims is *ranked, auditable evidence plus the test that would settle it* — which is what an analyst actually needs to act.

Specifically:

Component	What it really measures	What it does not mean
T temporal	The change started shortly before the metric broke	Precedence is necessary for causation, never sufficient
C cohort match	Row-level Jaccard between the change's declared blast radius and the affected cohort	A wide radius scores badly even if the change *was* the cause
D dose–response	Rank correlation of exposure against impact across sub-slices	**Uninformative for this incident** — both candidate radii fully contain the focal cohort, so exposure has no variance and every candidate correctly scores the neutral 0.5. The ranking here is carried by C and N, and we say so rather than manufacturing a number
N negative controls	Fraction of falsifiable predictions that held up	A passing control is a failure to falsify, not a confirmation
P learned prior	Beta-Bernoulli mean over past analyst verdicts	Starts flat at 0.5 and only sharpens ranking; it never gates

Three further things stated plainly:

- **The −$410k figure is an observed shortfall against a deseasonalized baseline**, not a causal impact estimate. There is no bootstrap interval behind it in this build, so none is shown.

- **Control rule 2 rests on a mechanism assumption.** The "segment siblings should also drop" control fires only for demand-side event types (campaign, price change, policy change, external, vendor incident) — see `SEGMENT_AGNOSTIC_EVENT_TYPES`. The reasoning: a regional demand shock has no mechanism by which it could spare Mid/SMB customers in that region, whereas a deploy targets a *code path*, and enterprise direct debit is a distinct code path. Without this gate the rule rejects the true cause as readily as the decoy. It is the single most load-bearing assumption in the system.
- **v1 flags drops only.** The generator's quarter-end multiplier is a known calendar effect the rolling-median baseline does not model, so a bidirectional detector would flag every quarter close. The correct fix is a calendar-regressor baseline (Round 2); for now the direction is restricted to the one the product cares about.

When it can't explain something, it says so. If no candidate clears the score floor, the card reports exactly that, lists which source systems are and are not connected, and recommends widening ingestion. The failure direction matters: a too-wide blast radius fails its controls, and a too-narrow one leaves nothing above the floor. The system degrades toward *I don't know*, not toward a confident wrong answer.

Run it

You need: Python 3.12 and about two minutes. **You do not need an API key** — the whole demo, and all 343 tests, run without one.

1. Install

```
git clone https://github.com/omega-sus67/LedgerLens.git
cd LedgerLens

# with uv (fastest)
uv venv --python 3.12
uv pip install --python .venv/bin/python -r requirements.txt

# or with plain Python, if you don't have uv
python3.12 -m venv .venv
.venv/bin/pip install -r requirements.txt
```

2. Build the data, then look at it

```
.venv/bin/python -m ledgerlens.gen_data     # ~10s. Writes data/ from SEED=20260815
.venv/bin/python -m ledgerlens.pipeline     # prints a full diagnosis to your terminal
```

`data/` is empty when you clone, and that is deliberate: it is deterministic generator output, so the repo carries the generator instead of the data. `gen_data` rebuilds every byte of it, identically, on any machine.

The second command is the fastest way to see what this does — a complete diagnosis card, evidence chain and recommended actions, printed as text.

3. Open the app

```
.venv/bin/python -m streamlit run app.py    # http://localhost:8501
```

4. Try these four things, in this order

The demo is built around one incident: renewals in DACH collapsed on 3 August. Everything below works out of the box at the default settings (`mrr_renewals` , as-of `2026-08-17`).

#	Do this	What to look for
1	Scroll to **Hypotheses** and find the red **REJECTED** card	A marketing campaign that is *more* temporally plausible than the true cause — and was killed anyway. Its **N** score sits at zero, and the pink row in its control table shows which check killed it
2	Scroll to **Recommended actions**	The `[P2]` line: the same campaign is innocent here but *guilty of something else*, with the query behind it
3	Switch **Persona** to `Growth Marketing` (sidebar)	The 🔒 banner. The numbers legitimately change, and the card names the policy that withheld the data rather than silently omitting it
4	Tick **Deploy source (github) not connected** (sidebar)	The system **refuses to name a cause**, says which feeds are missing, and tells you what to connect

Every number on screen has a query behind it. Expand 🔍 **show the queries behind this card** on any hypothesis to see the exact SQL and its logged result.

5. Optional: turn on the AI investigator

```
export GEMINI_API_KEY=...                    # free key from Google AI Studio works
.venv/bin/python -m streamlit run app.py
```

Then tick **Run the AI investigator** in the sidebar. It adds three LLM call sites — proposed checks, unverifiable causes, and persona-voiced prose — and **none of them can change a rank**. Roughly half a cent per diagnosis.

Provider-agnostic: `LEDGERLENS_LLM_PROVIDER=anthropic` switches vendor with no source change, and `LEDGERLENS_LLM_MODEL=...` swaps the model within one. See **LLM versus non-LLM** below and `docs/ai_decisions.md` .

Verify it yourself

```
.venv/bin/python -m pytest -q                # 343 tests, no API key needed
```

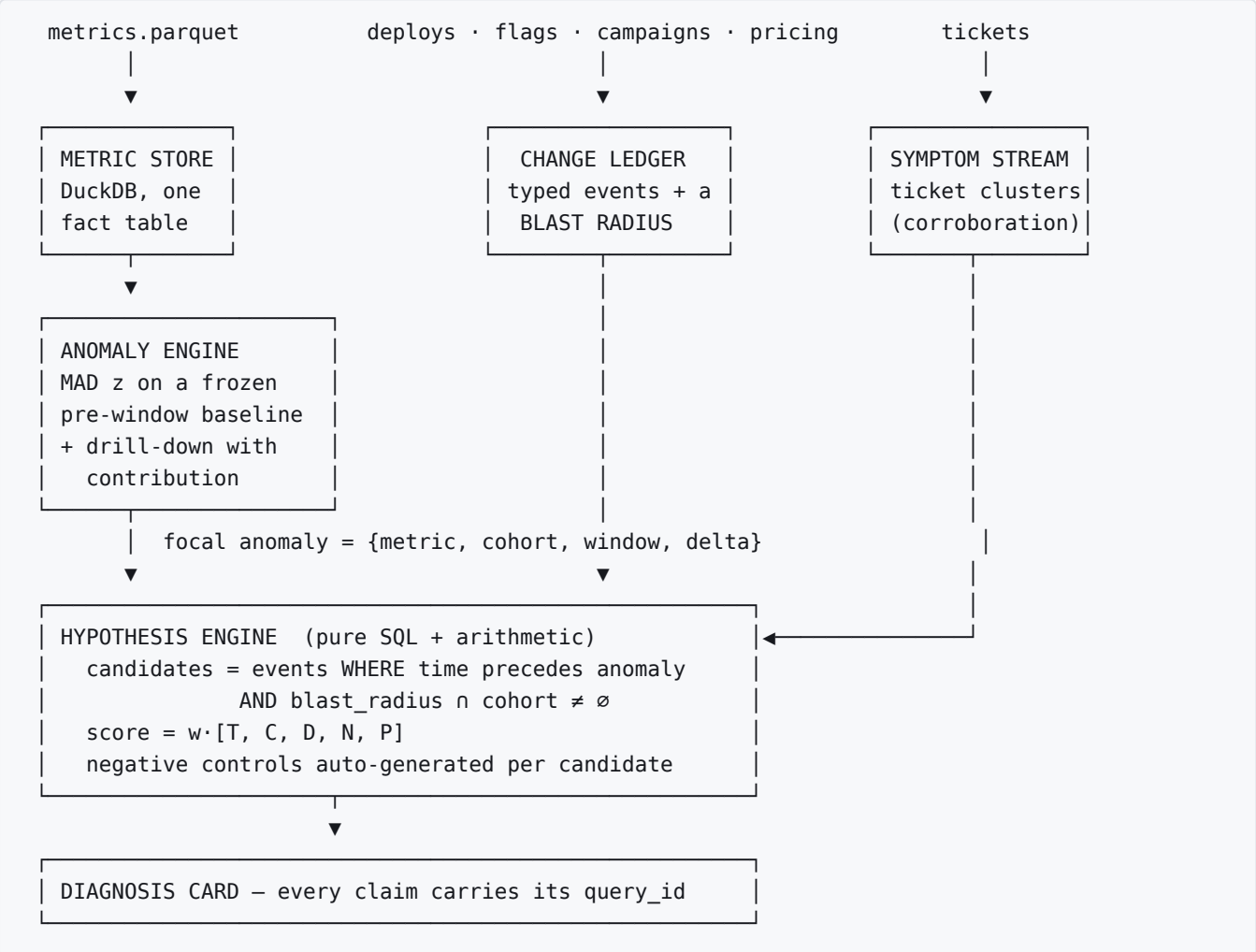
The suite is the claim. `tests/test_pipeline.py` walks a finished card, collects every `query_id` on it, and asserts each one still reproduces its logged output — so "every number is auditable" is enforced rather than asserted.

If something goes wrong

Symptom	Fix
`yaml` import error, or odd import failures	A ROS install is polluting `PYTHONPATH`. Prefix commands with `env -u PYTHONPATH -u VIRTUAL_ENV`
`IO Error: Could not set lock on file ... duckdb`	Another Streamlit or pytest process holds the database. Close it — the lock is exclusive
Empty `data/` or "file not found"	Run `python -m ledgerlens.gen_data` first
AI checkbox greyed out	`GEMINI_API_KEY` is not visible to the server. Export it, then restart Streamlit — reloading the browser is not enough

New to the codebase? `study_guide.md` is an ordered path through every document and source file, with checkpoints. `docs/how_it_works.md` teaches the vocabulary from zero.

Pipeline



Every claim is a query. `Store.q` is the only path to the database. It hashes the SQL plus its bound parameters into a `query_id`, logs the statement and a result preview, and hands both back. Any figure that cannot cite a `query_id` is a bug — and the acceptance test walks a finished card, collects every id, and asserts each one still reproduces its logged output.

The core primitive: blast radius

Every deliberate or observable change becomes a typed event with a **blast radius** — a set of dimension predicates describing which slices of the business it could plausibly touch.

```
ChangeEvent(
    event_id="deploy_sepa_v214",
    event_type="deploy",
    ts_start=datetime(2026, 8, 3, 2, 0),
    source="github",
    blast_radius={"region": ["DACH"], "payment_rail": ["sepa"]},
)
```

The insight is that in a real enterprise this linkage is **already recorded** and does not need to be inferred: a deploy knows its rollout regions and rails, a feature flag knows its targeting rules, a campaign knows its geo, a support ticket knows its account and the account knows its segment. `ledgerlens/ledger/connectors.py` derives blast radius by a **declared mapping** from that metadata — never by inference over prose. In production these read deploy metadata, LaunchDarkly targeting rules and campaign geo settings instead of the synthetic JSON in `data/`; the mapping is the only thing that changes.

That single design choice turns *"is this change relevant to this anomaly?"* into a **set intersection** — deterministic, sub-millisecond, and explainable in one sentence — instead of a graph traversal or an embedding similarity search.

The omission rule matters as much as the mapping: a dimension the source does not constrain is left **out**, making it unconstrained. That is why the marketing campaign's radius is region-only, and why it loses.

Three KPIs, and one that refuses to answer

LedgerLens carries three KPIs across three source systems on different cadences:

KPI	Unit	Aggregation	Source	Cadence	History
`mrr_renewals`	USD/day	sum	CRM	daily batch, 02:00 UTC	full
`new_logo_bookings`	USD/day	sum	CRM	daily batch, 02:00 UTC	full
`payment_success_rate`	rate	**ratio**	PSP webhook	every 15 minutes	**56 days**

The third one is the interesting one, and it is deliberately broken in a specific way.

It is too young to detect on. 56 days of history against a 120-day warmup, so `detect()` returns `None` before it looks at a single value. The card does not render a blank success box — it says what it lacks and asks for a window:

> **Insufficient history for automatic detection.** `payment_success_rate` launched 2026-06-23 — 56 days of history against a 120-day warmup. Detection is declined rather than run on a baseline that cannot support it.

Supply the window and the full chain still runs: the same SEPA connector release is found through a second KPI at `C = 1.00`, both negative controls pass, and support tickets corroborate at 10× lift. **Declining to detect is not declining to help.**

It is a rate, and rates are not additive. `fact_metric` has one `value` column, so the KPI is stored as two additive metrics — `payment_successes` and `payment_attempts` — and the contract declares `agg="ratio"`. `Store.series()` then issues `SUM(numerator)/SUM(denominator)`: a *weighted* rate, where a slice with 4,000 attempts counts more than one with 4. An unweighted average across slices would be a different and wrong number.

Two consequences we state rather than hide:

- **The drill-down is switched off for ratio KPIs.** Contribution analysis assumes a parent's delta is the sum of its children's. A rate is a mix effect plus a within-slice effect, and separating them needs a method this build does not have.
- **No seasonality is claimed.** The KPI has no prior August, so the card says *"no seasonal adjustment is applied"* rather than reporting a confident 0.0%.

It also gets its own alerting thresholds — a rate living at 98.2% cannot fall 3%, so the global gate would make it undetectable in principle. Full reasoning in `docs/sparse_kpi_decisions.md`.

Two audiences, identical evidence

Dashboards fail different readers differently: an analyst-shaped card is useless to a CFO, and a CFO-shaped card is useless to the engineer who has twenty minutes to stop the bleeding. The usual fix is to hand the card to an LLM and ask it to rewrite per audience — which puts a generative model between the evidence and the reader.

LedgerLens renders four personas from **one computation**. `pipeline.diagnose()` produces the evidence; `narrate(payload, persona)` renders it. Persona is accepted only by the narrator, which sits downstream of every `Store.q()` call — so it *cannot* reach a query.

Persona	Reads it to	Decision rights
Revenue Analyst (default)	route the work and audit every claim	all levers
CFO	decide what to tell the board	`hold_forecast`
Payments On-Call	stop the bleeding in twenty minutes	`rollback_release`, `disable_flag`
Growth Marketing	find out whether the budget cut is the problem	`restore_campaign_budget`

Decision rights are mechanical, not decorative. Every recommendation is bound to a controllable *lever*. A persona that does not hold that lever sees an escalation instead of an instruction:

> **On-call** — `[P0]` Roll back or hotfix `deploy_sepa_v214` for DACH · sepa… **CFO** — `[P0]` Escalate to the team owning the service: roll back or hotfix *a code release to the payment path* for DACH · sepa…

Note the CFO card never says `deploy_sepa_v214`. It never says a sha anywhere, including inside an escalation — `show_event_ids` governs action text as well as prose.

Actions carry the brief's full chain, and `models.py` lists the fields in that order:

```
driver -> controllable lever -> action -> expected impact -> owner -> confidence -> monitoring pl
```

`confidence` is the score of the *evidence the action rests on* — the hypothesis score for cause-linked actions, `1.0` for directly measured ones — not a probability that the action will work. The UI says so in as many words, and a test asserts the sentence is on the page.

The claim, machine-checked. `test_personas_differ_in_prose_but_share_every_query_id` asserts four distinct summaries against one identical 19-element `query_id` list. Full reasoning in `docs/persona_decisions.md`.

Stated precisely: identical evidence *at a fixed entitlement*. Persona changes prose and never a number. **Role** is a different axis and does change numbers — see the next section. Analyst, CFO and on-call share a role's worth of access and therefore share every `query_id`; Growth is entitled to less, and their card says so.

Abstention is the one thing that never varies by audience: a CFO is never handed a confident answer the analyst was refused.

Role-based entitlement: redaction that names itself

A KPI contract declares who may see which *cuts* of it. `mrr_renewals` carries one rule:

```
AccessRule(
    policy_id="fin.rail_detail",
    role="growth",
    hidden_dims=["payment_rail"],
    reason="Payment-rail revenue splits are finance-restricted; growth sees "
           "region and segment cuts only.",
)
```

Selecting the **Growth Marketing** persona removes `payment_rail` from the dimensions handed to the drill-down. The consequence is real and visible:

	Revenue Analyst	Growth Marketing
focal cohort	`DACH · Enterprise · sepa`	`DACH · Enterprise · A`
headline	**−85.2%, −$416,144**	**−34.6%, −$207,545**
top candidate	`deploy_sepa_v214` @ 0.700	`deploy_sepa_v214` @ 0.627
ranked order	—	**identical**
decoy rejected	`campaign_dach_cut`	`campaign_dach_cut`

Growth's number is smaller because their cut is shallower, not because the engine failed — and the card says exactly that, naming the policy and quoting its declared reason. `policy_id` and `reason` are read off the contract, never retyped into the narrator, so changing the contract changes the card; a test asserts it.

Two properties worth stating, both tested:

- **Entitlement hides cuts, not candidates.** The ranked set and its order are identical across roles. Scores *do* move (0.700 → 0.627) because the focal cohort legitimately changed — a policy that silently promoted a different cause would be a security hole dressed as a feature, so the ordering is asserted.
- **A redaction notice never computes what it hides.** The banner names the dimension, the policy and the reason. It does *not* count the withheld slices — that count would require running the unrestricted drill, i.e. computing the very answer the reader is not entitled to.

Scope, honestly: this is **dimension-level** entitlement. It is not row-level security, not measure-level, and not authentication — the role comes from the persona selector, not a login. Full reasoning and the named gaps in `docs/roles_decisions.md`.

When it doesn't know, it says so

The strongest thing this system does is refuse. Two different refusals, both reachable from the sidebar:

1. It declines to detect. `payment_success_rate` launched 2026-06-23 and has less history than its warmup requires. Rather than run a baseline that cannot support a verdict, detection declines and asks for a window — then explains the window you give it, with the uncertainty stated.

2. It declines to explain. Tick **"Deploy source (github) not connected"** in the sidebar. Every github-sourced change is removed at candidate generation — as if the connector had never been wired up. The anomaly is still real and still −85%, but the change that caused it was never in the ledger:

> `mrr_renewals` in DACH · Enterprise · sepa down 85% — **no connected change explains it**
>
> The anomaly is real … The closest candidate scored **0.38, below the 0.45 floor**. No recorded change whose blast radius touches this cohort clears the confidence floor. Connected sources: feature flags (launchdarkly), campaigns (calendar), pricing (pricing_db), support tickets (zendesk). **Not connected: deploys (github) — simulated as disconnected for this run**; vendor status feeds; billing policy change logs. The cause may well sit in one of those.
>
> `[P1]` data platform: **Connect github** so the next incident of this shape has candidates to test.

Three properties worth noting, all tested:

- **The cause is absent, not demoted.** `deploy_sepa_v214` appears nowhere — not ranked, not rejected, not in the evidence. An unconnected system produces no rows, not a low-scoring candidate; filtering anywhere later would model "we saw it and dismissed it", which is a different and less honest scenario.
- **The decoy does not win by default.** Removing github removes eight other deploys too, thinning the field — and `campaign_dach_cut` is *still* rejected by its own segment-sibling control. A decoy promoted the moment its rivals vanish would mean the control had never been doing the work.
- **The connectivity list is read off the KPI contract's lineage**, not retyped into the narrator. Add a connector to the contract and the card mentions it; a test fails if it does not.

Abstention never varies by audience: a CFO is not handed a confident answer the analyst was refused. Full reasoning in `docs/abstention_decisions.md`.

The feedback loop: a prior you can delete

The fifth score component, **P**, is the only number an analyst can move. Each hypothesis card carries 👍 / 👎 controls; a verdict is one row in `verdict`, and the prior is a Beta–Bernoulli posterior **re-counted from those rows on every diagnosis**.

There is no model state. Nothing is trained, nothing drifts, and **deleting a row puts the prior back exactly where it was** — `test_the_prior_is_derived_from_rows_not_kept_as_state` asserts precisely that. It is the cheapest possible learning mechanism that is still honest about what it learned and reversible when it learns wrong.

Three properties make it safe to ship:

- **It sharpens a ranking; it never decides one.** P is weighted `0.05`, so its entire range moves a total by at most 0.05. A candidate ten points below `SCORE_FLOOR` cannot be confirmed into confidence — there is a test that saturates the prior with 200 confirmations and checks the candidate *still* fails to clear the floor.
- **It cannot rescue what a control killed.** A decisive control failure zeroes N outright, and no amount of past agreement outvotes it.
- **It is auditable like everything else.** The prior goes through `Store.q`, so P carries a replayable `query_id` and the card states the evidence behind it — *"a Beta–Bernoulli posterior over 3 confirmed / 1 rejected past verdicts."* Before this, P was the one number on screen a reader could not open.

Feedback is offered to **every persona**. A verdict is not a lever: `decision_rights` gates actions, and the CFO who watched the forecast recover is as entitled to answer *"did this turn out to be right?"* as the analyst.

LLM versus non-LLM, and what a diagnosis costs

Nothing on the ranking path calls a model. Detection, attribution, candidate generation, scoring and negative controls are deterministic Python and SQL. That is a design decision, not an omission — and the check is that the entire test suite and the whole demo run with no API key set.

On top of that sits the investigator lane, and the split is exact:

	Who does it	What it may touch
detection, drill-down, cohort intersection, T/C/D/N/P, negative controls, decoy rejection	**deterministic SQL + Python**	the verdict
proposing *additional* checks from a fixed template vocabulary	**LLM proposes → this engine executes in SQL**	the card, never the score
listing causes that connected data cannot test	**LLM**	a separately-labelled panel
writing the headline and summary for one reader	**LLM, behind a numbers guard**	prose only

The boundary is enforced in code rather than by convention. Proposed checks are constructed with `decisive=False` — the field `controls.score_n` reads to zero out N — and are never passed to it. `test_the_lane_cannot_change_a_single_score` asserts that every score is byte-identical with the lane on and off.

Three guards, each of which reports what it caught. Proposals naming a dimension value, metric or template that does not exist are rejected *before* becoming a query, and the count is shown ("4 accepted, 2 rejected by validation"). The narrator is given every figure it may use, and any number in its prose that was not in that corpus discards the narration wholesale in favour of the deterministic template — the page says so when it happens. A vendor outage is reported as an outage, never as "nothing found".

The ⏱ **Telemetry** panel puts the accounting on the page:

stage	ms	share
drill	799.0	**62%**
rank	390.1	30%
detect	71.6	6%
seasonal	16.8	1%
symptoms	15.4	1%
narrate	0.9	0%
total	**1,293 ms**	

	count
registered queries executed	**89**
…served from cache	41
distinct `query_id`s replayable on the card	**22**
LLM calls / tokens / cost	**0 / 0 / $0.0000**

Two of those numbers deserve their distinction. **89** is what the diagnosis cost to produce; **22** is how much of it a reader can audit. Reporting the smaller one as "queries" would understate the work by roughly 4×, in the one panel whose whole job is honest cost accounting — so both are shown, under names that cannot be confused. Metadata lookups (`dim_universe`, `events`) are counted in neither: they carry no user-facing number and therefore no `query_id`.

The zero, priced. That row is the *default* state, with the investigator lane switched off. Turned on, it makes three calls per diagnosis on `gemini-2.5-flash` — roughly 6k input and 1.2k output tokens, about **$0.0048** at $0.30 / $2.50 per MTok (`GEMINI_API_KEY`). Switching to `claude-sonnet-5` via `LEDGERLENS_LLM_PROVIDER=anthropic` costs about **$0.0240** at $2.00 / $10.00 per MTok and requires no source change. Either way it adds checks and rewrites prose, and changes **none of the numbers above**.

Telemetry is one of exactly **two** numbers on the page that carry no `query_id` — the other is the redaction notice. Both are facts about the *process* rather than the data, and the UI says so rather than leaving a gap to be noticed. Full reasoning in `docs/telemetry_decisions.md`.

What the demo shows

The synthetic data has known ground truth, so the ranking can be *proven* right rather than merely look plausible.

- **True cause:** `deploy_sepa_v214` — a SEPA connector release on Aug 3 that collapses renewals for `DACH × Enterprise × sepa` to 15% of baseline. Aggregate impact −8.2%.
- **Decoy #1:** `campaign_dach_cut` — a DACH marketing budget cut two days earlier. Temporally *more* plausible than most candidates, region-overlapping, and completely innocent of this anomaly.
- **Decoy #2:** `pricing_us_q3` — eliminated by empty cohort intersection before it can be scored at all.

The result:

	T	C	D	N	total	
`deploy_sepa_v214`	1.00	**0.333**	0.50	1.00	**0.700**	4/4 controls pass
`deploy_dunning_v3`	0.26	0.091	0.50	1.00	0.443	
`deploy_billing_ui_v9`	0.14	0.030	0.50	1.00	0.393	
`flag_sepa_retry_beta`	0.00	0.111	0.50	1.00	0.383	enabled *after* onset
`campaign_dach_cut`	0.72	0.143	0.50	**0.00**	0.322	**REJECTED**

How the decoy dies. Its blast radius covers all of DACH — 21 slices — against a 3-slice anomaly, so **C** is 0.143 versus the deploy's 0.333. Then the control finishes it: *if this were a DACH-wide demand shock, DACH Mid and SMB renewals on the same rail should have dropped too.* They came in at **−1.3%**, flat. That is a decisive failure, **N** goes to zero, and the hypothesis is rejected outright rather than merely outranked.

And the honest second finding: the campaign *did* cause something real. Its objective-mismatch control shows DACH `new_logo_bookings` down **−31%** — exactly what a budget cut is supposed to do. It is the wrong metric for this anomaly, not a harmless event, and the card raises it as a separate P2.

Design notes worth knowing

Two baselines, deliberately. `scan_for_onset` uses a cheap trailing rolling median to find *where* it broke; `evaluate` fits a Theil–Sen model on the pre-window only and freezes it across the anomaly window to measure *how big* it is. Using the trailing median to measure would be a real bug: by the end of a 14-day window half the trailing 28 days are themselves anomalous, the baseline sags toward the new regime, and the reported delta shrinks from −8.2% to about −6.6%.

The pre-window must exclude the anomaly. If median and MAD are taken over the window itself, every day in it is ~85% low, the residual median moves with them, and z collapses toward zero — the anomaly quietly defines its own normal. `test_anomaly.py::test_mad_must_come_from_the_pre_window` asserts exactly this contrast.

Detection is advisory, not gating. `pipeline.run(cohort=..., window=...)` bypasses the detector entirely and runs the same downstream chain. Every detection blind spot — slow drifts, ratio metrics, interaction effects, offsetting regional moves that cancel at the root — becomes "we don't auto-surface this" rather than "we can't diagnose this."

Multiple testing. Benjamini–Hochberg runs across each drill-down level at q=0.10 and is recorded per node, but it never filters. Sibling slices are subsets of one parent and therefore positively dependent, which makes BH's guarantee approximate here; it is a principled sanity check on branch selection, and the negative controls — which need no distributional assumption — carry the actual rigor.

Symptoms corroborate; they never score. The 42-ticket `ERR_SEPA_504` spike attaches to the hypothesis as evidence but is excluded from the rubric, because its cohort fit measures the same signal **C** already does and scoring both would double-count it.

Determinism. `SEED = 20260815` throughout, and the generator solves its two scaling constants in closed form against realized window sums rather than hand-tuning an RNG — so the aggregate delta lands on −8.17% by construction, not by luck.

Layout

```
config.py                    constants, weights, thresholds, SEED
app.py                       Streamlit UI
ledgerlens/
  models.py                  Pydantic contracts + the cohort algebra
  store.py                   DuckDB lifecycle + query registry
  gen_data.py                synthetic generator (writes ground_truth.json)
  anomaly.py                 detection, measurement, drill-down, BH
  ledger/connectors.py       deterministic event ingestion
  ledger/symptoms.py         ticket clustering
  hypothesis.py              candidates + five-component scoring
  controls.py                negative control generation and evaluation
  learning.py                Beta-Bernoulli priors
  narrate.py                 template narrator + guarded LLM narration
  llm.py                     provider seam: gemini + anthropic adapters
  investigator.py            the LLM lane: proposed checks, unverified causes, guard
  pipeline.py                orchestration
tests/                       343 tests; test_pipeline.py is the acceptance test
docs/
  how_it_works.md            start here if the vocabulary is new
  ai_decisions.md            the LLM lane, end to end
  *_decisions.md             why each subsystem is built the way it is
  design/                    the original build contract + architecture rationale
  taskflow/                  the plans these were built from, kept as history
```

Deviations from `docs/design/IMPLEMENTATION_SPEC.md` are marked `# SPEC-GAP:` at the point of departure, with the reason. The substantive ones are the three control-rule fixes described above and the practical-significance gate (`MIN_ABS_DELTA_PCT`) that makes the "no false flag before the incident" test deterministic rather than a coin flip.

Not built in this cut

Per the spec's own MVP path: `effect.py` (diff-in-differences with bootstrap CI), `ambiguity.py` (the discriminating test), the confirm/reject learning loop in the UI, and the bounded `explore` pass.

`ledger/normalizer.py` — the LLM event normalizer — is deliberately cut rather than merely unbuilt. It is the one call site of the four that is *not* additive: the spec has it multiply a hypothesis score by an LLM-emitted extraction confidence, which puts a model on the ranking path and contradicts the property every other part of this design protects. Its schema (`models.ExtractedSignal`) and fields (`ChangeEvent.extraction` , `.confidence`) stay declared as the honest record of a deferred decision. Reasoning in `docs/ai_decisions.md` D2.